An All-Obtuse Convex Hexagon Compiler for Planar Shifts of Finite Type

Exact decoding under unrestricted contacts

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Abstract

We give an effective construction that maps every nearest-neighbor $\mathbb{Z}^2$-shift of finite type S to a finite protoset $P(S)$ of unmarked strictly convex rational hexagons. Every interior angle of every output hexagon lies in $(\pi/2, \pi)$. Tiles may be translated, rotated, or reflected, and contacts are unrestricted. Even under this permissive model, every $P(S)$-tiling is forced to be edge-to-edge, its abstract cell complex is the standard honeycomb, and it locally decodes to a point of S. Conversely, every point of S deforms the affine honeycomb into a $P(S)$-tiling. Hence $P(S)$ tiles the plane if and only if S is nonempty.

The construction yields undecidability of tileability for finite sets of strictly convex rational hexagons whose angles are all obtuse, with all Euclidean isometries and arbitrary contacts allowed. If S is strongly aperiodic, then $P(S)$ is aperiodic in the same unrestricted shape-only sense. As an explicit instance, the phase-two compilation of the 13 Kari-Culik Wang tiles has 4,956 pairwise noncongruent prototiles. An exact finite artifact enumerates its states, graph edges, and coronas and checks the finite inequalities used in the construction.

1. Introduction

Matching rules are easy to state when tiles carry colors, arrows, or signed edges. They are harder to enforce when the final objects are unmarked polygons, reflections are permitted, and one cannot assume an edge-to-edge tiling. A geometric encoding must first control arbitrary contacts, then recover the intended local state at every vertex, and finally prove that no other global assembly remains.

This paper gives such an encoding for any nearest-neighbor $\mathbb{Z}^2$-shift of finite type. The output polygons have a deliberately restricted form:

1. every tile is a strictly convex hexagon;
2. every coordinate is rational;
3. every interior angle is strictly obtuse;
4. every Euclidean isometry, including reflection, is permitted; and
5. no contact rule is imposed beyond disjoint interiors and coverage of the plane.

The all-obtuse condition removes T-junctions. A finite phase carrier fixes the three honeycomb edge directions. Locally legal source triples are collared at honeycomb vertices. A mixed-radix horizontal displacement assigns a different length to every compatible pair of collared endpoint states. At an arbitrary physical vertex, the three decoded graph edges form a triangle in a line graph. Bipartiteness forces that triangle to be a star, recovering one collared state rather than three pairwise compatible fragments.

The main result is a compiler theorem, rather than a theorem restricted to aperiodic inputs. Aperiodicity and undecidability follow as separate corollaries.

Theorem 1.1. All-obtuse compiler

For every nearest-neighbor $\mathbb{Z}^2$-shift of finite type S, one can effectively construct a finite set $P(S)$ of unmarked strictly convex rational hexagons such that:

1. every angle of every polygon in $P(S)$ lies in $(\pi/2, \pi)$;
2. $P(S)$ tiles $\mathbb{R}^2$ if and only if $S \neq \varnothing$;
3. every unrestricted $P(S)$-tiling is edge-to-edge and trivalent, with abstract cell complex the standard honeycomb;
4. its decoded phase map is the universal cover of H_2 and its face symbols form a configuration in S;
5. up to one global orthogonal map and translation, that tiling is exactly the bounded piecewise-affine deformation associated with its decoded source point; and
6. every configuration in S gives such an unrestricted $P(S)$-tiling.

Here unrestricted means that translations, rotations, and reflections are allowed and no edge-to-edge convention, selected atlas, matching relation, or one-sidedness assumption is supplied.

Corollary 1.2. Undecidability

It is undecidable whether a finite set of unmarked strictly convex rational hexagons, all of whose angles lie in $(\pi/2, \pi)$, tiles the plane when all Euclidean isometries and unrestricted contacts are allowed.

Corollary 1.3. Strong aperiodicity

If S is nonempty and every point of S has no nonzero $\mathbb{Z}^2$-period, then $P(S)$ is aperiodic: it tiles the plane and no admitted $P(S)$-tiling has a nonzero translational symmetry.

The priority boundary is narrow. Robinson announced a reduction to finite sets of convex lattice-coordinate pentagons or hexagons in 1971, including rotations and reflections, but the cited paper does not give that construction or its proof. The contribution here is an explicit compiler with the all-obtuse output restriction and a classification of every unrestricted-contact tiling. We make no claim to have originated the broad idea that convex polygons can carry undecidable tiling systems.

2. Source systems and polygon tilings

2.1. Nearest-neighbor shifts

Let A be a finite alphabet and let

$$R_E, R_N \subseteq A \times A$$

be directed east and north adjacency relations. Define

$$S = S(A, R_E, R_N)$$

to be the set of all $x \in A^{\mathbb{Z}^2}$ satisfying

$$(x_{i,j}, x_{i+1,j}) \in R_E, \qquad (x_{i,j}, x_{i,j+1}) \in R_N$$

for every $(i, j) \in \mathbb{Z}^2$.

We place these symbols on the faces of a honeycomb. Two of the three face-neighbor directions carry R_E and R_N. The third direction is unconstrained. This is an embedding of a square nearest-neighbor shift into the triangular face lattice, not a claim about arbitrary three-relation triangular-lattice shifts.

A point $x \in S$ is periodic if some nonzero $(u, v) \in \mathbb{Z}^2$ satisfies

$$x_{i+u,j+v} = x_{i,j}$$

for all i, j. We call S strongly aperiodic when it is nonempty and no point of S is periodic in this sense.

2.2. Shape-only tilings

A prototile is an unmarked simple polygon considered up to Euclidean congruence. A P-tiling is a covering of $\mathbb{R}^2$ by congruent copies of members of a finite protoset P, with pairwise disjoint interiors. Reflections are allowed. Contacts may be partial, and an edge-to-edge convention is not assumed. A finite protoset is allowed to be empty; the empty protoset is a valid no-instance and does not tile.

This definition is the only placement rule used below. Symbols, phase coordinates, and graph states are construction and proof devices. They are not retained as marks on the polygons.

3. The finite phase carrier

3.1. A four-face honeycomb torus

Let H_2 have vertices

$$B_{ij}, W_{ij}, \qquad i, j \in \mathbb{Z}/2\mathbb{Z}.$$

The three neighbors of B_{ij} are

$$W_{ij}, \qquad W_{i-1,j}, \qquad W_{i,j-1}.$$

Its four faces are the simple cycles

$$C_{ij} = (B_{ij}, W_{ij}, B_{i+1,j}, W_{i+1,j-1}, \\ B_{i+1,j-1}, W_{i,j-1}).$$

Thus H_2 has 8 vertices, 12 edges, and 4 faces. It is a torus cellulation, and its universal cover is the standard honeycomb.

Here is a direct check. Modulo two, the three displayed neighbors of every black vertex are distinct. Each C_{ij} is a simple six-cycle; every edge occurs in exactly two such cycles, with opposite induced orientations; and the incident face sectors form one cyclic link at each vertex. The resulting complex is therefore a connected closed orientable surface. Its Euler characteristic is

$$8 - 12 + 4 = 0,$$

so it is a torus. If the indices i, j are instead allowed to range over $\mathbb{Z}$, the same face cycles embed as the affine honeycomb of Section 4.1. Quotienting that planar complex by the translations $(i, j) \mapsto (i+2, j)$ and $(i, j) \mapsto (i, j+2)$ gives H_2. Hence the planar honeycomb is its universal cover.

The modulus two phase has two jobs. It makes every face boundary a simple six-cycle, and it ensures that the endpoints of a state-graph edge determine one fixed phase edge rather than several parallel witnesses.

3.2. Collared vertex states

Place a source face x_{ij} at every $(i, j) \in \mathbb{Z}^2$. At a black honeycomb vertex use the ordered triple

$$B_{ij}(x_{ij}, x_{i-1,j+1}, x_{i-1,j}),$$

and at a white vertex use

$$W_{ij}(x_{ij}, x_{i,j+1}, x_{i-1,j+1}).$$

A black triple $B_{ij}(a, b, c)$ is locally legal when

$$(c, a) \in R_E, \qquad (c, b) \in R_N.$$

A white triple $W_{ij}(a, d, b)$ is locally legal when

$$(a, d) \in R_N, \qquad (b, d) \in R_E.$$

A vertex state is a phase vertex of H_2 together with one such locally legal ordered triple.

Order all black states and all white states effectively:

$$Q_B = \{q_0^B, \ldots, q_{B-1}^B\}, \qquad Q_W = \{q_0^W, \ldots, q_{W-1}^W\}.$$

If either set is empty, the construction returns the empty protoset. This case cannot occur when S is nonempty.

3.3. Legal coronas

Name a central source face a and its six cyclic neighbors

$$b, d, e, f, g, c.$$

The seven symbols form a legal corona when

$$(c, a), (a, e), (b, d), (g, f) \in R_E$$

and

$$(g, a), (a, d), (c, b), (f, e) \in R_N.$$

For phase C_{ij}, associate the six states

$$\begin{aligned} q_0 &= B_{ij}(a, b, c), \\ q_1 &= W_{ij}(a, d, b), \\ q_2 &= B_{i+1,j}(e, d, a), \\ q_3 &= W_{i+1,j-1}(f, e, a), \\ q_4 &= B_{i+1,j-1}(f, a, g), \\ q_5 &= W_{i,j-1}(g, a, c). \end{aligned}$$

Let G be the simple bipartite graph with vertex set

$$Q_B \sqcup Q_W.$$

An unordered pair $\{q, r\}$ is an edge of G when q, r occur consecutively in at least one legal corona cycle. Multiple witnesses create one graph edge.

Every state already contains its phase vertex. Hence a graph edge has a fixed black phase endpoint, a fixed white phase endpoint, and one fixed black-to-white phase vector. The graph has no loops or parallel edges.

4. Exact polygon geometry

4.1. The affine base honeycomb

Use the centrally symmetric hexagon

$$\begin{aligned} v_0 &= (2, 0), & v_1 &= (1, 3/2), & v_2 &= (-1, 3/2), \\ v_3 &= (-2, 0), & v_4 &= (-1, -3/2), & v_5 &= (1, -3/2). \end{aligned}$$

Its face-center lattice has basis

$$a = (-3, 3/2), \qquad b = (0, 3).$$

The face at (i, j) is the translate of the base face by

$$ia + jb.$$

These translates form an affine image of the regular honeycomb.

4.2. Mixed-radix displacements

Put

$$D = 4096BW.$$

Assign a horizontal displacement to every state:

$$z(q) = \begin{cases} b/D, & q = q_b^B, \\ B(w+1)/D, & q = q_w^W. \end{cases}$$

For a legal corona cycle $(q_0, \dots, q_5)$, define the polygon

$$P(q_0, \dots, q_5) = \mathrm{poly}\big(v_0 + (z(q_0), 0), \dots, v_5 + (z(q_5), 0)\big).$$

Finally, $P(S)$ is the finite set of congruence classes of these polygons over all four phases and all legal coronas.

Every coordinate belongs to $(2D)^{-1}\mathbb{Z}$. Multiplication by $2D$ gives an integer-coordinate representative. The construction is finite and effective from A, R_E, R_N.

4.3. Every graph edge has a unique length

Let $q_b^B q_w^W$ be an edge of G, and put

$$\delta = z(q_w^W) - z(q_b^B) = \frac{B(w+1) - b}{D}.$$

Its numerator belongs to the block

$$\{Bw + 1, \dots, Bw + B\}.$$

These blocks are pairwise disjoint, so δ determines both b and w. Moreover,

$$0 < \delta \le \eta, \qquad \eta = \frac{1}{4096}.$$

The squared lengths are

$$L_h(\delta) = 4 + 4\delta + \delta^2$$

for a horizontal phase edge and

$$L_s(\delta) = \frac{13}{4} - 2\delta + \delta^2$$

for either slant direction. The function L_h is strictly increasing on $(0, \eta]$, and L_s is strictly decreasing there. Their ranges are disjoint because

$$L_s(\delta) < \frac{13}{4} < 4 < L_h(\delta).$$

A state pair fixes one edge of the simple phase graph H_2, including the sign of the slant component of its black-to-white vector. It cannot occur in both slant directions. It follows that distinct graph edges have distinct lengths.

Lemma 4.1. Corona cycles are pairwise noncongruent

No two distinct legal corona cycles define congruent polygons.

Proof A congruence preserves the cyclic word of side lengths up to rotation and reversal. Every length identifies one graph edge. Consecutive distinct edges in a simple graph recover their unique common endpoint. The cyclic edge word therefore reconstructs the cyclic state word. Two congruent polygons came from the same corona cycle up to the dihedral presentation already absorbed by congruence. □

4.4. Quantitative convexity and obtuseness

The following calculation supplies the deformation margin used later.

Lemma 4.2. Uniform all-obtuse bound

For any legal corona and every assignment produced above, the polygon is strictly convex and every interior angle is obtuse. More precisely, every cross product of consecutive forward edges is at least

$$3 - 3\eta > 0,$$

and every dot product of consecutive forward edges is at least

$$\frac{5}{4} > 0.$$

Equivalently, the dot product of the two rays from a polygon vertex to its neighbors is at most $-5/4$.

Proof Write $\delta_k \in (0, \eta]$ for the black-to-white displacement difference on side k, with the index attached to the corresponding graph edge. In cyclic order the six forward edge vectors have the forms

$$\begin{aligned} e_0 &= (-1 + \delta_0, 3/2), \\ e_1 &= (-2 - \delta_1, 0), \\ e_2 &= (-1 + \delta_2, -3/2), \\ e_3 &= (1 - \delta_3, -3/2), \\ e_4 &= (2 + \delta_4, 0), \\ e_5 &= (1 - \delta_5, 3/2). \end{aligned}$$

Their consecutive cross products are of two types:

$$\frac{3}{2}(2 + \delta_k)$$

and

$$3 - \frac{3}{2}(\delta_k + \delta_{k+1}).$$

They are at least $3 - 3\eta$.

Their consecutive forward dot products are of two types:

$$(2 + \delta_k)(1 - \delta_{k+1})$$

and

$$\frac{9}{4} - (1 - \delta_k)(1 - \delta_{k+1}).$$

The first is at least $2(1-\eta) > 5/4$, and the second is at least $5/4$. Positive cross products give strict convexity. At a vertex the two interior rays are $-e_{k-1}$ and e_k, whose dot product is the negative of the displayed forward dot product. Thus every interior angle lies in $(\pi/2, \pi)$. □

The same calculation applies to the three rays at a honeycomb vertex. In their cyclic order, their pairwise cross products have the two forms just displayed. Their cyclic order therefore survives every permitted displacement.

5. Arbitrary contacts become a honeycomb

Lemma 5.1. All admitted tilings are edge-to-edge and trivalent

Let P be any finite set of strictly convex polygons whose interior angles all lie in $(\pi/2, \pi)$. Every unrestricted P-tiling is edge-to-edge, and exactly three tile corners meet at every tiling vertex.

Proof First note that the tiling is locally finite. Let $r > 0$ be the minimum inradius and $M < \infty$ the maximum diameter among the finitely many prototiles. Every tile meeting a compact set K has a disjoint radius-r inball contained in the bounded M-neighborhood of K. Area packing permits only finitely many such inballs.

Consider a point x in the relative interior of a side of one tile. That tile contributes a sector of angle π. All sectors contributed by tiles on the other side must sum to π.

If a tile corner occurred at x, one corner would leave a positive angle smaller than $\pi/2$, which cannot be filled by another permitted corner or by an edge-interior sector of angle π. Two corners would already sum to more than π. Hence no tile corner can lie in the relative interior of another tile side. The remaining half-plane is occupied by exactly one edge-interior sector.

A change from one neighboring side to another along a side would create a tile corner in its relative interior. Thus every side is paired along its full length with one whole side.

At a tiling vertex, only polygon corners occur. Two corners sum to less than 2π, while four corners sum to more than 2π. Exactly three corners meet. □

Lemma 5.2. Honeycomb regularization

The abstract cell complex of any unrestricted $P(S)$-tiling is the standard $\{6, 3\}$ honeycomb.

Proof Lemma 5.1 makes the tiling a locally finite edge-to-edge trivalent hexangulation.

Replace every physical face by a unit regular hexagon and glue corresponding full sides. The topological realization of this abstract complex is homeomorphic to the original polygon tiling and hence to $\mathbb{R}^2$. At every vertex, three sectors of angle $2\pi/3$ meet. The regularized surface is therefore locally Euclidean, including at its vertices.

The regularized path metric is complete. Triangulate every physical polygon and its regular replacement by joining its barycenter to its six vertices. On each prototile class, the resulting cellwise map and its inverse are piecewise affine and Lipschitz. There are only finitely many prototile classes, so one constant works for every triangle. The cellwise maps agree on glued edges and give a global bi-Lipschitz homeomorphism from the physical Euclidean plane to the regularized intrinsic surface. Completeness of the physical plane therefore implies completeness of the regularized metric.

Choose an oriented frame in one face and analytically continue it across edges. Since the surface is simply connected and flat, this gives a developing local isometry

$$\mathrm{dev} : M \longrightarrow \mathbb{R}^2.$$

A local isometry from a complete connected Euclidean surface to $\mathbb{R}^2$ has the path-lifting property for complete geodesics and is a covering map. Because $\mathbb{R}^2$ is simply connected, dev is a global Euclidean isometry. Its regular hexagonal faces form the standard honeycomb. □

The completeness step rules out a developing map whose image is only a proper open subset of the plane. The covering step then supplies injectivity, not merely local flatness.

6. Decoding every shape-only tiling

Lemma 6.1. The line-graph star lemma

At every vertex of an arbitrary $P(S)$-tiling, the three incident physical sides recover three graph edges of G with one common endpoint state.

Proof Side length identifies a graph edge. Each pair among the three physical sides occurs as a corner of one of the three incident polygon faces. The corona that defined that polygon shows that the corresponding pair of graph edges shares an endpoint.

The three graph edges are distinct. Otherwise an incident face would contain two equal consecutive graph edges, contradicting the simplicity of its phase face cycle.

Thus the three graph edges form a triangle in the line graph $L(G)$. In a simple graph, three pairwise incident edges either share one common vertex or form a triangle. The graph G is bipartite, so it has no triangle. The three edges form a star at one state q. □

Lemma 6.2. A physical edge recovers opposite state endpoints

If a physical side decodes to $\{q, r\} \in E(G)$, its two physical endpoints decode to q and r, in one of the two orders.

Proof Choose either incident polygon. In its corona cycle, the two ends of the side are q and r. At the physical endpoint corresponding to q, the two consecutive graph edges of that polygon share

q. Two distinct edges in a simple graph have at most one common endpoint, so the common star state from Lemma 6.1 is q. The other endpoint similarly recovers r. A reflected placement may exchange the physical order, but cannot identify the two states. □

Lemma 6.3. The phase map is the universal honeycomb cover

The decoded state labels define a cellular covering from the physical honeycomb to H_2.

Proof Map each physical vertex to the phase vertex contained in its decoded state, each physical edge to the phase edge determined by its graph edge, and each physical face to the phase face contained in its corona cycle.

At a physical vertex, the three incident graph edges project to distinct phase edges. If two projected to the same phase edge, the physical face between them would map two consecutive boundary edges to one edge of the simple cycle C_{ij}. Thus the vertex star maps bijectively to a phase vertex star.

It remains to exclude folding along an edge. Suppose the two physical faces adjacent to one edge mapped to the same phase face. At either endpoint, the two other physical rays belonging to those faces would then map to the same adjacent boundary edge of that phase face. This contradicts injectivity of the vertex-star map. Hence the two sides of every physical edge map to the two different incident phase faces.

The cellular map is now a local homeomorphism in face interiors, edge interiors, and vertex stars.

For completeness, the image is all of H_2. Starting from any image vertex, bijectivity of its physical vertex star supplies all three incident phase edges and their opposite endpoints. Iterating through the connected one-skeleton reaches every phase vertex and edge. Along each lifted edge, the no-folding argument supplies its two distinct incident phase faces, so every face is reached as well.

Now take barycentric subdivisions of the physical complex and H_2. A vertex in either subdivision represents a face, an edge, or a vertex of the original complex. Face recognition, the two-sided edge result, and the bijective vertex-star result say exactly that the link of each physical barycentric vertex maps isomorphically to the corresponding target link. Consequently, the inverse image of every open barycentric star is a disjoint union of open barycentric stars, each mapped homeomorphically onto the target star. This is the cellular covering criterion. The phase map is therefore a covering of H_2. By Lemma 5.2 its domain is simply connected, so it is the universal cover. □

The local orientation sign of a covering between oriented surfaces is a locally constant function with values in $\{1, -1\}$. The physical honeycomb is connected, so this sign is constant. Reflections are therefore handled by one global choice rather than a hidden local chirality rule.

Lemma 6.4. Polygon placements have one global orthogonal frame

There is one $R \in O(2)$ such that the orthogonal part of every tile placement is R.

Proof A graph edge has one fixed canonical black-to-white vector because its state endpoints include their phase vertices. The two canonical phase faces incident to that edge traverse this vector in opposite directions and lie on opposite sides of its line.

Consider two physical polygons sharing the corresponding side. After orienting the canonical vector consistently, their orthogonal parts both map it to the same physical vector. Two orthogonal maps agreeing on a nonzero vector are either equal or differ by reflection in its line. In the second case, the two canonical face interiors, initially on opposite sides, are placed on the same physical side of the shared edge. Their interiors overlap near the edge. This is forbidden in a tiling.

The orthogonal parts agree across every adjacency. The honeycomb adjacency graph is connected, so one R applies to every tile. □

Proposition 6.5. Source reconstruction

Every unrestricted $P(S)$-tiling determines a point of S, up to a global deck-coordinate choice and the global frame R.

Proof Lemma 6.3 identifies the physical cell complex with the universal cover of H_2. Use its deck coordinates to index physical faces by $(i, j) \in \mathbb{Z}^2$. The six states around a physical face came from one legal corona and therefore give the same central symbol. Lemma 6.2 makes these state assignments agree across physical edges.

The phase edge specifies whether an adjacency is east-west, north-south, or the unconstrained third direction. The legal triples and coronas impose R_E and R_N on the first two directions. The reconstructed symbols therefore define $x \in S$. Every step used only side lengths, polygon incidence, and the unmarked phase geometry encoded by those lengths.

Fix one physical vertex and its canonical lifted phase vertex. Lemma 6.4 gives one orthogonal part R, and Lemma 6.2 gives the canonical vector across every incident edge. Induction through the connected honeycomb now shows that every physical vertex has the form

$$R\big(u + (z(q(u)), 0)\big) + t$$

for one translation t. Thus the arbitrary tiling is exactly the deformation associated with the reconstructed source point, up to R and t. □

Proposition 6.5 proves the soundness half of Theorem 1.1. It also classifies every component of the shape-admitted tiling space. No selected generator orbit is substituted for the complete space.

7. Constructing a tiling from every source point

The converse requires a global deformation statement. A large finite patch does not suffice.

Lemma 7.1. Quantitative honeycomb deformation

Let $x \in S$. Move each vertex of the affine honeycomb horizontally by the displacement of its collared state. Add the barycenter of every face, move it by the average of the six incident vertex displacements, and extend the map affinely over the six center-fan triangles. The resulting map

$$F_x : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

is a proper orientation-preserving local homeomorphism at bounded distance from the identity.

Proof Every honeycomb vertex receives one state determined by the adjacent source faces, so neighboring faces move a shared vertex to the same point. Every vertex and barycenter displacement belongs to $[0, \eta] \times \{0\}$.

For each center-fan triangle, direct expansion of its determinant gives

$$\det(F_x(v_k) - F_x(c), F_x(v_{k+1}) - F_x(c)) \geq 3 - 3\eta > 0.$$

The ray-star calculation following Lemma 4.2 gives the same lower bound for every consecutive pair among the three deformed edge rays at a honeycomb vertex. Thus every face fan stays embedded and positively oriented, while every deformed three-face star keeps cyclic degree one. These facts make the piecewise-affine extension a local homeomorphism in triangle interiors, across triangulation edges, across tile edges, at face centers, and at honeycomb vertices.

On every triangle, $F_x - \mathrm{id}$ is a convex combination of its vertex displacements. Hence

$$\|F_x(y) - y\| \leq \eta$$

for every $y \in \mathbb{R}^2$. If K is compact, then

$$F_x^{-1}(K) \subseteq \{y : \mathrm{dist}(y, K) \leq \eta\}.$$

The preimage is closed and bounded, hence compact. Thus F_x is proper. □

Proposition 7.2. Completeness of the compiler

If $S \neq \varnothing$, then $P(S)$ tiles $\mathbb{R}^2$.

Proof Choose $x \in S$. A proper local homeomorphism from the plane to the plane has open image. Properness also makes its image closed. The image is nonempty, so connectedness of $\mathbb{R}^2$ makes it surjective. It is a covering map, and the simply connected target has no nontrivial connected cover. Hence F_x is a homeomorphism.

The images under F_x of the affine honeycomb faces cover the plane with disjoint interiors. Each image face is exactly the polygon associated with its legal corona. These images form a $P(S)$-tiling. □

Combining Proposition 7.2 with Proposition 6.5 gives

$$P(S) \text{ tiles } \mathbb{R}^2 \quad \Longleftrightarrow \quad S \neq \varnothing.$$

This proves the tileability and decoding parts of Theorem 1.1.

8. Consequences

8.1. Undecidable tileability

Nonemptiness of nearest-neighbor $\mathbb{Z}^2$-shifts of finite type is undecidable. The construction of $P(S)$ uses finite enumeration and rational arithmetic. Therefore an algorithm deciding unrestricted

tileability for finite sets of strictly convex rational all-obtuse hexagons would decide whether S is nonempty. This proves Corollary 1.2.

The reduction permits rotations and reflections and does not assume edge-to-edge contacts. The latter property is a conclusion of Lemma 5.1. The decision-problem encoding admits the empty finite list, as specified in Section 2.2. After a common multiplication by $2D$, every output coordinate is an integer. The number of polygons is not bounded independently of the source.

8.2. Strong aperiodicity

Let S be strongly aperiodic and suppose a $P(S)$-tiling $\mathcal{T}$ had translational period $t \neq 0$. The regularization and decoding are local and commute with translations. Thus t preserves the regular affine honeycomb obtained from $\mathcal{T}$. It is a nonzero honeycomb lattice vector and induces a nonzero element of $\mathbb{Z}^2$ in the deck coordinates of Lemma 6.3. Proposition 6.5 then gives a point of S with that period, a contradiction. This proves Corollary 1.3.

8.3. From ordinary aperiodicity to strong aperiodicity for a planar SFT

For completeness, suppose a two-dimensional SFT has a configuration with one nonzero period vector p. Quotienting by p gives a nonempty one-dimensional SFT over a finite alphabet of transverse strips. Every nonempty one-dimensional SFT contains a periodic point, obtained from a directed cycle in its finite presentation. Lifting that point gives a doubly periodic configuration of the original SFT. Consequently, a Wang shift with no doubly periodic configuration has no singly periodic configuration either.

9. An explicit Kari-Culik instance

9.1. The 13 source tiles

Write a Wang tile as (S, W, N, E). The six division tiles, with multiplier $1/3$, are

S	W	N	E
1	0/3	0	1/3
1	1/3	0	2/3
1	2/3	1	0/3
2	0/3	0	2/3
2	1/3	1	0/3
2	2/3	1	1/3

The seven doubling tiles, with multiplier 2, are

S	W	N	E
0	-1	$0'$	-1
0	0	$0'$	0
0	0	1	-1
1	-1	1	0
1	-1	2	-1
1	0	2	0
$0'$	0	1	-1

The colors 0 and $0'$ are distinct. The symbols $0/3, 1/3, 2/3$ are formal colors.

9.2. An exact bi-infinite source point

Set

$$\alpha_0 = \sqrt{2}$$

and define

$$\alpha_{j+1} = \begin{cases} 2\alpha_j, & \alpha_j < 1, \\ \alpha_j/3, & \alpha_j > 1. \end{cases}$$

The orbit stays in $[1/3, 2)$. Its inverse is

$$\alpha_{j-1} = \begin{cases} 3\alpha_j, & \alpha_j < 2/3, \\ \alpha_j/2, & \alpha_j > 2/3. \end{cases}$$

Every α_j is a nonzero rational multiple of $\sqrt{2}$, so the orbit never meets a rational branch boundary. The recurrence is therefore unambiguous for every $j \in \mathbb{Z}$.

For $n, j \in \mathbb{Z}$, put

$$A_j(n) = \lfloor n\alpha_j \rfloor$$

and let

$$\lambda_j = \frac{\alpha_{j+1}}{\alpha_j} \in \{2, 1/3\}.$$

Define

$$\begin{aligned} S_{n,j} &= A_j(n) - A_j(n-1), \\ N_{n,j} &= A_{j+1}(n) - A_{j+1}(n-1), \\ W_{n,j} &= \lambda_j A_j(n-1) - A_{j+1}(n-1), \\ E_{n,j} &= \lambda_j A_j(n) - A_{j+1}(n). \end{aligned}$$

On a division row, retain W and E in formal thirds notation. A zero on the south side of row j is labeled $0'$ exactly when

$$\alpha_{j-1} \in [1/3, 1/2],$$

and a zero on the north side is labeled $0'$ exactly when

$$\alpha_j \in [1/3, 1/2].$$

All other numerical zeros are labeled 0.

Lemma 9.1. Exact floor classification

For every $(n, j) \in \mathbb{Z}^2$, the four labels above form exactly one of the 13 Kari-Culik tiles.

Proof Fix a row and write $\alpha = \alpha_j$, $\beta = \alpha_{j+1}$.

First suppose $\alpha < 1$, so $\beta = 2\alpha$. For an integer k, write

$$k\alpha = A + t, \qquad A \in \mathbb{Z}, \quad 0 \le t < 1.$$

Then

$$2A - \lfloor 2k\alpha \rfloor = -\lfloor 2t \rfloor \in \{-1, 0\}.$$

It follows that $W, E \in \{-1, 0\}$, while $S \in \{0, 1\}$ and $N \in \{0, 1, 2\}$. The definitions give the exact identity

$$2S + W = N + E.$$

These ranges and this identity leave precisely the six numerical patterns

$$(0, -1, 0, -1), \quad (0, 0, 0, 0), \quad (0, 0, 1, -1),$$
$$(1, -1, 1, 0), \quad (1, -1, 2, -1), \quad (1, 0, 2, 0).$$

If $N = 0$, then $\beta < 1$, hence $\alpha \in [1/3, 1/2)$; the north zero is therefore $0'$. If $(S, W, N, E) = (0, 0, 1, -1)$, the south zero may occur on either side of $\alpha = 2/3$. Indeed, the inverse branch is 3α below $2/3$ and $\alpha/2$ above $2/3$, so a south zero is $0'$ exactly when $\alpha > 2/3$. This gives the two distinct table entries $(0, 0, 1, -1)$ and $(0', 0, 1, -1)$. The remaining south zeros in the two $N = 0$ patterns are unprimed because there $\alpha < 1/2$. Thus the seven doubling tiles are exactly the possible formal patterns.

Now suppose $\alpha > 1$, so $\beta = \alpha/3$. For an integer k, write

$$\lfloor k\alpha \rfloor = 3\lfloor k\alpha/3 \rfloor + r_k, \qquad r_k \in \{0, 1, 2\}.$$

Then $W = r_{n-1}/3$ and $E = r_n/3$ as formal thirds. Also

$$S \in \{1, 2\}, \qquad N \in \{0, 1\},$$

and

$$\frac{1}{3}S + W = N + E.$$

Equivalently, if $W = w/3$ and $E = e/3$,

$$S + w = 3N + e.$$

The stated ranges leave exactly the six division rows in the table. A north zero on a division row is unprimed because $\alpha > 1$. This proves exhaustion and uniqueness. □

Lemma 9.2. Matching and occurrence

The floor construction is a valid bi-infinite Kari-Culik configuration. All 13 base tile types occur.

Proof Horizontal matching is an identity:

$$E_{n,j} = W_{n+1,j}.$$

Vertical matching follows from

$$N_{n,j} = S_{n,j+1}.$$

The prime convention is also identical on the shared vertical edge: both sides use the condition $\alpha_j \in [1/3, 1/2]$. Lemma 9.1 supplies one table tile at every lattice site, including negative n and negative j.

For the final assertion, write

$$\alpha_j = q_j\sqrt{2}$$

and abbreviate

$$(a, b; c, d) = (A_j(n-1), A_j(n); A_{j+1}(n-1), A_{j+1}(n)).$$

Direct exact evaluation gives one occurrence of every type:

Tile	(n, j)	q_j	$(a, b; c, d)$	Resulting (S, W, N, E)
1	$(-1, 0)$	1	$(-3, -2; -1, -1)$	$(1, 0/3, 0, 1/3)$
2	$(2, 0)$	1	$(1, 2; 0, 0)$	$(1, 1/3, 0, 2/3)$
3	$(-2, -3)$	3/4	$(-4, -3; -2, -1)$	$(1, 2/3, 1, 0/3)$
4	$(-2, 3)$	4/3	$(-6, -4; -2, -2)$	$(2, 0/3, 0, 2/3)$
5	$(-2, 0)$	1	$(-5, -3; -2, -1)$	$(2, 1/3, 1, 0/3)$
6	$(3, 0)$	1	$(2, 4; 0, 1)$	$(2, 2/3, 1, 1/3)$
7	$(-3, -2)$	1/4	$(-2, -2; -3, -3)$	$(0, -1, 0', -1)$
8	$(4, -2)$	1/4	$(1, 1; 2, 2)$	$(0, 0, 0', 0)$
9	$(-1, 1)$	1/3	$(-1, -1; -2, -1)$	$(0, 0, 1, -1)$
10	$(-1, -1)$	1/2	$(-2, -1; -3, -2)$	$(1, -1, 1, 0)$
11	$(-2, -1)$	1/2	$(-3, -2; -5, -3)$	$(1, -1, 2, -1)$
12	$(-2, 2)$	2/3	$(-3, -2; -6, -4)$	$(1, 0, 2, 0)$
13	$(-3, -1)$	1/2	$(-3, -3; -6, -5)$	$(0', 0, 1, -1)$

For $\alpha_j = (p/q)\sqrt{2}$ and $k > 0$, every entry is certified by

$$A_j(k) = \left\lfloor \frac{\lfloor \sqrt{2k^2p^2} \rfloor}{q} \right\rfloor, \qquad A_j(-k) = -A_j(k) - 1.$$

The second identity uses irrationality of $k\alpha_j$. The needed row coefficients are

$$q_{-3} = \frac{3}{4}, \quad q_{-2} = \frac{1}{4}, \quad q_{-1} = \frac{1}{2}, \quad q_0 = 1, \quad q_1 = \frac{1}{3}, \quad q_2 = \frac{2}{3}, \quad q_3 = \frac{4}{3}.$$

This finite table proves occurrence. It is not being used in place of the symbolic all-site classification in Lemma 9.1. □

Culik proves that this Wang set admits no periodic tiling. The one-period-to-two-period argument in Section 8.3 makes its full Wang shift strongly aperiodic. Corollary 1.3 therefore applies.

9.3. Size of the compiled instance

For the phase-two Kari-Culik compilation, exact enumeration gives

quantity	value
source symbols	13
black vertex states B	568
white vertex states W	600
state-graph edges	3632
legal coronas	4956
congruence classes	4956
D	1395916800.

Thus the explicit shape-only protoset contains 4,956 pairwise noncongruent strictly convex all-obtuse rational hexagons.

10. Exact artifact

The retained artifact is

```
workbench/kc-convex-candidate.mjs
```

It uses `BigInt` arithmetic for source floors, displacement coordinates, cross products, dot products, squared lengths, and multiplier identities. For a chosen winding modulus and phase modulus it enumerates every source symbol, collared vertex state, graph edge, and legal corona. For the unlifted Kari-Culik instance, replay:

```
node workbench/kc-convex-candidate.mjs 1 2 100
```

The principal output is:

```
alphabetSize                      13
blackCount                       568
whiteCount                       600
graphEdgeCount                  3632
coronaCount                     4956
congruenceClassCount            4956
exactConvexityChecks           29736
exactVectorConsistencyChecks    3632
exactUniqueLengthChecks         3632
coordinateFingerprintSha256
9be30b636e6a103c9360b5ffc7043e7d255df1a7e317da5369d86eb8c79e5fbd
```

The artifact checks:

1. the 13 multiplier identities in exact thirds units;

2. the exact bi-infinite source formulas on the requested finite audit window;
3. every face cycle is simple;
4. every exact corner cross product is positive;
5. every exact interior-ray dot product is negative;
6. every graph edge has one canonical black-to-white vector;
7. every graph edge has one squared length;
8. different graph edges have different squared lengths; and
9. distinct corona cycles remain distinct under the dihedral action.

The SHA-256 value is a reproducibility fingerprint over the ordered scaled-integer coordinate enumeration. It is not an independent proof certificate. The symbolic arguments in Sections 4, 6, 7, and 9 prove the infinite statements. The finite artifact checks the consequential finite enumerations and catches implementation drift.

11. Effectivity and scope

The compiler is a direct finite algorithm:

1. enumerate locally legal black and white state triples;
2. enumerate seven-face legal coronas in each of four phases;
3. construct the simple compatibility graph G;
4. index its black and white vertices;
5. assign the mixed-radix rational displacements;
6. emit one exact hexagon per dihedral corona class.

The output cardinality depends on the source alphabet and its local language. This theorem supplies no fixed bound on the number of polygons. It is a many-polygon compiler and does not settle tileability for any fixed number of convex polygons.

If $a = |A|$, then the four phases give the coarse bounds

$$B \le 4a^3, \qquad W \le 4a^3, \qquad |P(S)| \le 4a^7.$$

These bounds are sufficient for effectivity and are not intended to be sharp. When $B = 0$ or $W = 0$, the algorithm returns the empty protoset before defining D.

The construction is also unsuitable as a practical ceramic inventory in its current form. Its edge differences may be far below ordinary manufacturing tolerances, and the Kari-Culik instance has thousands of prototiles. Those facts do not affect the exact tileability theorem, but they delimit its material interpretation.

12. Relation to prior work

Robinson's 1971 paper announces variable-cardinality undecidability for convex lattice-coordinate pentagons or hexagons even with rotations and reflections. It does not present that construction in the cited article. This prior announcement prevents a broad priority claim for convex-polygon undecidability.

Demaine and Langerman give a proved unrestricted-contact compiler using three nonconvex polygons. Stade gives a two-polygon compiler with a highly concave carrier. Yang and Zhang study translation-only tilings by orthogonally convex polyominoes and ask fixed-cardinality questions. These results have different output classes and cardinality goals. The result here isolates a direct all-obtuse convex-hexagon mechanism.

Convex hexagon tilings can have irregular and non-edge-to-edge behavior in other angle regimes. The present contact reduction uses the strict lower bound $\pi/2$ at every corner, rather than a general classification of convex hexagon tilings.

13. Limitations and proof boundary

The construction advances an output-class and forcing theorem. It does not claim a new symbolic hull when instantiated on a known SFT. A faithful nonconvex compiler applied to the same source may produce a locally equivalent symbolic presentation.

The explicit Kari-Culik instance is large and visually close to an affine honeycomb. Its role is to instantiate the theorem with exact data and a strongly aperiodic source.

The novelty search behind this draft is finite. Robinson's unpublished announcement is a reason to describe the result as an explicit all-obtuse construction and proof, without an unconditional historical priority claim.

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